Derivation appendix: 07 — the radiative decay of the muon in the soft-broken vacuum

one loop and Barr-Zee, against MEG II

1. Lepton couplings of every physical scalar

derived here — §1

Independent checks

- $Y^{(a_k)} = iY^{(r_k)}$, $Y^{(c_k)} = \sqrt{2}Y^{(r_k)}$ emerge from the Lagrangian
- Goldstone couplings flavour-diagonal at every soft vacuum
- all 7 states agree with feynlag's extractor at a non-aligned point

Would be falsified by: a missing $\sqrt{2}$ on the charged coupling (falsify run)

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2. One-loop neutral dipole in closed form, exact in m_f/m_ϕ

derived here — §2.1

Claim

$$c_L = \sum_f \frac{\frac{1}{2}I_1(t_f)(m_e Y_{ef} Y^*_{\mu f} + m_\mu Y^*_{fe} Y_{f\mu}) + I_2(t_f) m_f Y^*_{fe} Y^*_{\mu f}}{4 m_\mu m_\phi^2}$$

Independent checks

- equals [HKZ13] (A3) with external masses kept, to 10^{-4}, at a heavy and a light scalar
- $t \to 0$ gives the [OST16]/[HKZ13] logs; [OST16] (30)'s m_i/m_μ on the non-flip term is a typo

Would be falsified by: the [OST16] (30) non-flip factor (falsify run)

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3. Charged one loop $= -\frac{1}{2}\times$ neutral non-flip

imported — §2.2

Independent checks

- the [Lavoura03] (40)/(41) and [OST16] (32) ratios agree independently

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4. Barr–Zee fermion and W terms

imported — §2.3 ([OST16], [HKZ13], [Ilisie15])

Independent checks

- top term = [HKZ13] (14) to 1 percent
- W bracket = [Ilisie15] $F^{(3)}$ to 1 percent; [HKZ13]'s 0.11 is 14 percent lower (set C, not implemented)

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5. Barr–Zee $H^{\pm}$ term and the sign $\lambda = -g/v$

derived here — §2.4

Independent checks

- low-energy theorem: the [Ilisie15] vertices against feynlag's diphoton amplitude

Would be falsified by: $\lambda = +g/v$ (falsify run)

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6. Assembly and BR normalization

imported — §2.5

Independent checks

- the [HKZ13] $\mu \to e\gamma$ bound is reproduced after the W-term rescaling

Would be falsified by: a factor-2 normalization error (falsify run)

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7. $\mu \to e\gamma$ vanishes at the exact-S_3 vacuum (residual $\mathbb{Z}_2$)

derived here — §3–4

Independent checks

- $\mathcal{B} < 10^{-24}$ (round-off) at 40 exact points $\times$ 3 values of μ_3
- $\mathcal{B} \propto (60^\circ - \varphi)^2$ off the alignment
- identical at the S_3 image (30 points, all configurations)

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8. Quark couplings along the vacuum direction

assumed — §5

Independent checks

- same ansatz as notebook 06's $\kappa_t = \kappa_V$
